

KISANCALL + AGROCHAIN

Complete SIH 2026 Product, Market & Technical Research Report

Voice-first procurement coordination + verifiable transaction layer

Know the price. Know your slot. Know your queue. Know your payment. Have proof of what happened.

Prepared for SIH 2026 | Based on the team's KisanCall project document, AgroChain concept, and current 2026 public sources.

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Executive verdict

The combined project is technically buildable and has strong SIH potential, but only if it is positioned as a focused procurement-access and coordination platform rather than a generic agriculture app. KisanCall should remain the farmer-facing layer; AgroChain should remain a narrow trust/proof layer underneath it.

The strongest product story is: the government already has digital procurement infrastructure and data; the farmer experience still has fragmentation, digital-literacy and last-mile communication gaps. KisanCall reduces the need for unnecessary travel and repeated status checking. AgroChain adds a tamper-evident proof of key procurement events.

Dimension	Assessment	Why it matters
SIH problem alignment	9.5/10	Directly maps to registration, slots, queues, notifications, procurement and payment tracking.
Technical feasibility	9/10	No core component is research-only; voice, web, mobile, database and blockchain are all buildable.
Market differentiation	8.5/10	Government and private systems already cover pieces, so differentiation must be the integrated farmer experience.
Demo potential	10/10	A real phone call + live queue + government price + blockchain proof creates a memorable demo.
Production readiness today	6.5/10	Production government integrations, compliance, language coverage and telecom scale still require partnerships.
Overall winning potential	9/10	Strong if scope is controlled and the demo proves one end-to-end journey flawlessly.

Source basis: the team's KisanCall project document defines the SIH requirement and existing architecture.

[T1] The current public procurement environment and government AI programs were checked separately.

[S1]-[S9]

1. Problem statement and real-world workflow

1.1 The official SIH problem

The supplied project document identifies SIH problem statement SIH26032 under the Department of Consumer Affairs. The requested capabilities are farmer registration and slot booking, real-time queue position, SMS/app notifications, procurement and payment status tracking, and lower congestion/waiting time at procurement centres. [T1]

1.2 What the problem really is

The real problem is not the absence of digital agriculture. India already has e-NAM, state procurement portals, central monitoring systems and AI-based farmer assistance. The problem is the last-mile experience: farmers often interact with multiple systems and a physical procurement centre, while the most useful status information is not presented as one simple, proactive, voice-accessible journey.

NITI Aayog's 2025 agriculture roadmap identifies six systemic barriers that are highly relevant here: siloed agricultural data systems, trust deficits, a persistent phygital divide, ecosystem fragmentation, talent gaps and constrained capital. [S7]

1.3 Real-world procurement workflow

A simplified cross-state workflow is:

Government / procurement agency announces procurement

|

v

Farmer registration / KYC / land-crop verification

|

v

Farmer receives centre / token / slot (varies by scheme/state)

|

v

Farmer travels to procurement centre

|

v

Gate entry / arrival registration

|

v

Queue + weighing + quality assessment

|

v

Procurement transaction / lot / bill entry

|

v

Payment processing

|

v

Farmer receives or checks payment status

This is intentionally described as a general workflow, not a claim that every state uses exactly the same screens or actors. e-NAM documents its own detailed digital journey, including registration, lot tracking from gate entry to gate exit and electronic payment; state MSP procurement flows can differ. [S2][S3]

1.4 What is already digitised

- e-NAM supports farmer registration through the portal, mobile app or mandi registration; farmers can view lots, confirm sale price and track lot progress from gate entry to gate exit. [S2][S3]
- The Central Foodgrains Procurement Portal publishes current-season registration, transactions, procurement, MSP payments, procurement centres and average-payment-time reporting. On 21 Aug 2026 it showed 51.28 lakh registered farmers and 33.96 lakh transacting farmers for RMS 2026-27 wheat. [S1]
- Government AI systems are also expanding: Kisan-eMitra is voice-enabled and supports farmer questions in 11 languages; Bharat-VISTAAR launched in 2026 as a voice-first multilingual agricultural DPI initiative. [S6][S8][S9]

1.5 The actual gap

The product gap is therefore best defined as: a farmer-facing, voice-first coordination layer that can sit above existing procurement data and systems, proactively tell the farmer what to do next, and optionally attach cryptographic proof to key transaction events.

1.6 Farmer pain points

Pain point	What the farmer needs	KisanCall response
Uncertain arrival time	A clear slot and useful arrival window	Slot confirmation + reminder call/SMS

Long physical waits	Knowing current queue and expected turn	Live queue + estimated wait
Schedule changes	Immediate notice without checking an app	Outbound call/SMS
Price uncertainty	Current mandi reference price with date/source	Government-reported mandi data
Payment uncertainty	Simple status in everyday language	AI payment-status explanation
Low smartphone comfort	No new digital workflow to learn	Voice-first access
Poor connectivity	A fallback channel	Voice/SMS; app is optional
Trust concerns	Evidence that a transaction was recorded	Transaction ID / blockchain proof for selected events

2. Market landscape and existing solutions

2.1 e-NAM

e-NAM is a major benchmark. Its farmer module allows registration, lot visibility, sale-price confirmation and step-by-step lot progress. Its farmer benefits include price/arrival information and electronic payments. The official site currently lists more than 1,400 integrated mandis, and the Government reported 1.89 crore farmers and 2.78 lakh traders registered by 30 June 2026. [S2][S3][S10]

2.2 Central Foodgrains Procurement Portal

CFPP is a government monitoring/reporting layer for foodgrain procurement. It exposes current-season registration, transaction, procurement, MSP paid, centres, agencies and average payment time. It is evidence that the system of record already contains valuable signals KisanCall could eventually consume through authorised integration. [S1]

2.3 Kisan-eMitra

Kisan-eMitra proves that voice-enabled AI for Indian farmers is not speculative. A 2025 PIB note said it had resolved over 95 lakh queries from 53 lakh farmers as of 15 July 2025, and its voice-enabled experience supports 11 major regional languages. Its scope, however, is primarily government scheme and farmer-service assistance rather than procurement-centre queue orchestration. [S6]

2.4 Bharat-VISTAAR

This is the most important new competitor/category signal. Bharat-VISTAAR was launched in Phase 1 in February 2026 as a nationwide, voice-first, multilingual agricultural digital public infrastructure initiative. It aims to integrate government agriculture data and scientific resources to provide weather, market-price, pest, soil, crop and government-service information through calls/chat and later app channels. [S8][S9]

This means KisanCall should NOT claim novelty simply because it uses voice AI or gives mandi prices. Those capabilities are becoming part of the public digital-agriculture stack. The novelty must be the procurement workflow and transaction-proof integration.

2.5 State-level WhatsApp booking experience

Andhra Pradesh's paddy procurement slot-booking initiative is a useful adoption case. Reporting on the programme described limited uptake and cited familiarity with walk-ins, smartphone limitations, digital hesitation and trust barriers, while digitally comfortable farmers benefited from faster processing and less waiting. [S11]

2.6 Market comparison

Capability	e-NAM	State procurement portals	CFPP	Kisan-eMitra	Bharat-VISTAAR	AgroChain	KisanCall + AgroChain
Farmer registration	Yes	Yes, varies	Monitoring	Scheme-oriented	Planned/linked	Yes, own platform	Yes
Slot / token coordination	Limited / scheme-dependent	Often yes	Reporting	No	Not core	No	Core
Live queue position	Not a core nationwide feature	State/centre dependent	No	No	No	No	Core
Outbound voice	No core procurement agent	Varies	No	Voice-enabled	Voice-first	No	Core
Mandi reference prices	Yes	Varies	No	Context dependent	Yes	Can use external data	Yes
Procurement status	Lot/transaction dependent	Yes in many systems	Yes	No	Not core	Can record	Core
Payment visibility	Yes in relevant workflows	Yes	Yes	Scheme payments	Government-service info	Can record	Core
Blockchain proof	No	No	No	No	No	Core	Core for selected events
Works without smartphone	Partially via mandi/assisted flows	Varies	N/A	Voice capability	Voice-first	Depends	Yes
Unified farmer journey	Partial	Fragmented by state/scheme	Government monitoring	Scheme support	Advisory/DPI	Trading/proof	Procurement-focused layer

3. What current systems are still missing

- A consistent procurement-centre experience that puts slot, queue, price and payment in one conversational flow.
- Proactive communication: the system calls the farmer before the farmer has to ask.
- A design that assumes the smartphone is optional rather than mandatory.
- A narrow, auditable proof layer for the agreed price/quantity/procurement/payment events, without putting personal data on-chain.
- A clear distinction between government-of-record data and the app's own coordination data.

3.1 What we should NOT claim

- Do not claim that farmers have no digital procurement systems.
- Do not claim e-NAM lacks electronic payments or lot tracking.
- Do not claim the AI voice idea is completely new; Kisan-eMitra and Bharat-VISTAAR prove otherwise.
- Do not claim direct government payment API integration unless an authorised integration actually exists.
- Do not describe Agmarknet daily market data as second-by-second live market pricing.

4. Combined KisanCall + AgroChain solution

4.1 Product definition

KisanCall is the farmer-facing voice and digital coordination layer. AgroChain is the trust/proof layer. The two should be integrated through backend events rather than merged into one monolithic application.

KisanCall tells the farmer what is happening. AgroChain proves what happened.

4.2 End-to-end farmer journey

1. Farmer registers by app or voice
2. System identifies preferred mandi and crop

3. Farmer receives an available procurement slot
4. AI call/SMS confirms slot and reference price information
5. Farmer arrives; staff marks ARRIVED
6. Queue engine computes position and ETA
7. Crop is weighed/verified; procurement record is created
8. Backend creates a proof record/hash for key transaction fields
9. Procurement status becomes COMPLETED
10. Payment status becomes PROCESSING / PAID
11. AI calls the farmer with the latest status
12. Farmer can request the transaction/proof reference

4.3 What AgroChain should store

Do not put the farmer's personal data, phone number, bank details, Aadhaar number or full transaction record on the public blockchain.

Data	Supabase	Blockchain
Farmer name / phone / profile	Yes	No
Slot / queue	Yes	No
Full payment details	Yes	No
Bank details	Yes, protected	No
Transaction ID / event ID	Yes	Yes
Hash of selected transaction payload	Yes	Yes
Agreed quantity / price hash	Yes	Yes
Procurement completion event	Yes	Yes
Payment event hash	Yes	Yes
Timestamp / block reference	Yes	Yes

5. Feature and competitor comparison

Feature	KisanCall + AgroChain	Typical procurement portal	e-NAM	Voice scheme bot
Registration	Voice + app	Portal/centre/app	Portal/app/mandi	Voice/chat
Slot booking	Core	State/scheme dependent	Not universal procurement slot layer	No
Live queue + ETA	Core	Usually centre-specific	Not a core feature	No
Proactive AI call	Core	SMS/notifications vary	Not core	Possible
Government mandi reference price	Yes	Varies	Yes	Yes in relevant context
Procurement tracking	Core	Yes	Lot lifecycle	No
Payment status	Core	Yes	e-payment support	Scheme-specific
Blockchain proof	Yes, selected events	No	No	No
Smartphone optional	Yes	Varies	Partially	Yes
Human fallback	Yes	Usually staff/centre	Mandi support	Scheme support
Single conversational journey	Yes	Usually fragmented	App-oriented	Scheme-oriented

6. Farmer usability and adoption strategy

The adoption strategy must assume that technology adoption is a product problem, not a farmer intelligence problem. The Andhra Pradesh slot-booking experience is a useful warning: digital tools can work technically and still see low adoption when users are unfamiliar with them or prefer an existing process. [S11]

Design rule	Implementation
Voice first	Farmer can complete core tasks with a phone call.
Zero learning curve	Use natural questions such as 'mera slot kab hai?' instead of menu-heavy IVR.
Hindi/English first	Start with two languages, then add regional languages.
Repeat important facts	Every slot/payment confirmation is spoken twice when appropriate

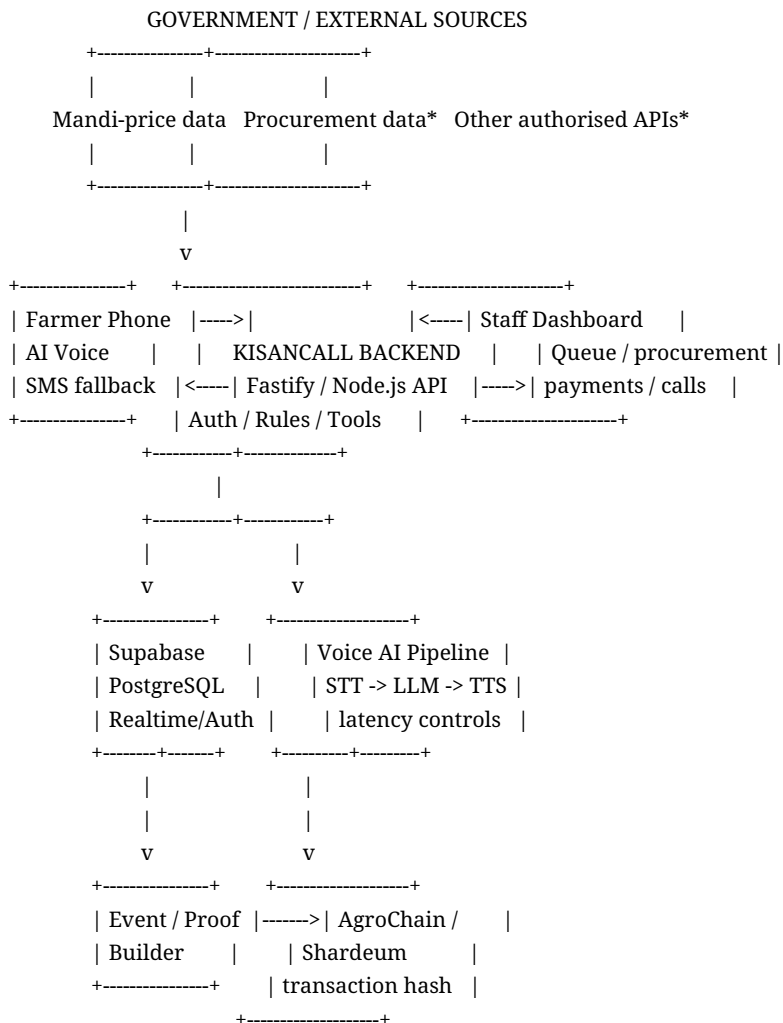
	and optionally followed by SMS.
No hallucinated answers	AI only responds from structured backend tools/data sources.
Fallback	After recognition failures, use keypad or human escalation.
Trust	State source and timestamp for prices; expose transaction proof after procurement.
Assisted registration	Staff/centre can create or correct farmer records.
Simple app	Large text, few screens, local terminology, low-data design.
Status words	Use plain statuses: BOOKED, ARRIVED, IN QUEUE, PROCURED, PAYMENT PROCESSING, PAID.

7. Complete technical architecture

7.1 Non-technical explanation

Think of the system as one brain with several ways to reach it. The mobile app, farmer web dashboard and phone call all ask the same backend for information. The staff dashboard writes operational updates. Supabase stores live operational data. Government sources provide externally verified market/procurement information. AgroChain records only selected proof events.

7.2 ASCII architecture



*For SIH, use public/authorised data sources or simulated status updates where direct government integration is not available. Do not claim production access that you do not have.

7.3 Recommended stack

Layer	Recommended technology	Reason	SIH status
Farmer mobile app	Expo + React Native	One codebase; fast iteration	Use
Farmer web dashboard	Next.js	Good for simple status view	Use if time allows
Staff dashboard	Next.js	Tables, realtime queue, role-based UI	Use
Backend	Fastify + Node.js	Fast, lightweight, team familiarity	Use
Database	Supabase Postgres	Auth, realtime, relational data	Use
Realtime	Supabase Realtime / WebSockets	Live queue updates	Use
Voice telephony	Twilio for prototyping, India-compliant provider for production	Fast API integration; India restrictions must be respected	Prototype only
STT	Deepgram	Multilingual/real-time speech stack	Use / benchmark
LLM	Groq-hosted open model or low-latency equivalent	Fast tool-calling for short responses	Use / benchmark
TTS	Deepgram Aura / Flux TTS or alternative	Low-latency speech	Use / benchmark
Knowledge / FAQs	Prefer structured tools; Qdrant only if needed	Avoid unnecessary RAG	Optional
Blockchain	AgroChain + Shardeum	Existing project / proof layer	Use narrowly
Hosting	Vercel for web; Render or equivalent for backend	Fast deployment	Use
Monitoring	Sentry / platform logs	Catch production-like failures	Optional but recommended

8. Dashboards and application architecture

8.1 Farmer mobile app

- Register / login
- Choose crop and mandi
- Book or confirm slot
- View slot, queue, ETA
- View current government-reported price with date/time
- View procurement status
- View payment status
- View proof/transaction ID
- Call KisanCall

8.2 Farmer web dashboard

Keep this intentionally simpler than the staff dashboard. It is a fallback for smartphone/web users.

8.3 Staff dashboard

- Today's farmer list
- Pending registrations
- Slot capacity
- Arrivals
- Live queue
- Estimated waiting time
- Procurement processing
- Payment status updates
- Call logs and failed calls
- Unreachable farmer list

- Manual override / correction with audit trail
- Blockchain proof reference

8.4 Roles

Role	Permissions
Farmer	Own profile, slot, queue, procurement and payment status
Operator	Queue, arrival, procurement entry, call actions
Supervisor	All operator functions + overrides + reports
Admin	Centre configuration, users, integrations, audit logs

9. Data, APIs and external dependencies

This is the area where the prototype must be honest. The public data layer is feasible today; direct government operational integrations are a later partnership step.

Data / API	What we need	Source / method	Prototype	Production
Mandi prices	Commodity, market, state, min/max/modal price, date	data.gov.in AGMARKNET dataset/API	Yes	Yes, subject to API policy/access
Farmer identity/profile	Name, phone, centre, crop	Our database; future authorised Farmer ID/AgriStack integration	Yes	Authorised integration
Slot capacity	Centre capacity, date/time, current count	Our database configured by staff	Yes	Yes
Queue state	Arrived/served counts, timestamps	Our realtime DB	Yes	Yes
Procurement status	Booked/arrived/verified/procured	Staff dashboard or authorised source	Yes	API/integration
Payment status	Pending/processing/paid	Staff/admin in prototype; authorised govt source in production	Yes	Authorised integration
Centre master data	Name, code, address, working hours	Government/open data + operator setup	Yes	Verified govt master
Weather (optional)	Forecast for farmer convenience	Government/approved API	Optional	Optional
SMS	Slot/payment notifications	Indian SMS provider / approved route	Yes	Yes
Telephony	Inbound/outbound call	Telephony provider	Yes	India-compliant carrier/cloud telephony
Blockchain	Transaction proof/hash	Shardeum RPC / AgroChain contract	Yes	Yes, with governance

9.1 Mandi-price source: important correction

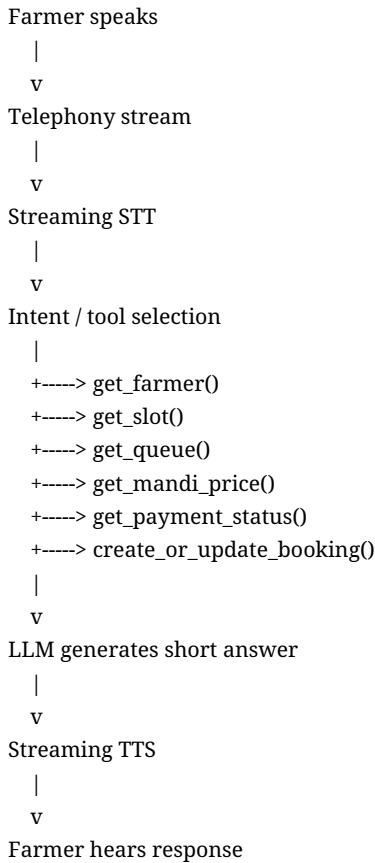
The current daily-price dataset on the Open Government Data platform is generated through AGMARKNET and contains wholesale maximum, minimum and modal prices on a daily basis. The dataset page was updated in 2026. Therefore, in the product UI say 'latest government-reported daily mandi price' rather than implying exchange-like tick-by-tick pricing. [S12]

9.2 data.gov.in access

The OGD platform documents API-key generation for registered users and provides a Data API layer. [S13][S14] Recommended implementation: server-side API call -> cache -> normalize -> timestamp -> expose a small internal endpoint to the voice agent.

10. Voice AI engineering

10.1 Pipeline



10.2 Context design

Do not place the entire farmer record, long FAQ documents and full history into every prompt. Use a compact structured context object, for example:

```

farmer_id
language
preferred_mandi
crop
today_slot
queue_position
queue_eta
latest_price
price_date
procurement_status
payment_status
last_call_outcome
  
```

This is consistent with the team's original KisanCall design: give the model only the exact facts needed for the current call rather than a huge context window. [T1]

10.3 Latency budget

Stage	Target	How
Voice capture / streaming	Continuous	Stream audio; detect speech end quickly

STT	~100-400 ms incremental	Use streaming STT / turn detection
Backend tool call	~50-250 ms internal	Local API + indexed DB query
LLM time-to-first-token	~100-400 ms target	Small fast model; concise prompts; tool calling
TTS first audio	~100-400 ms target	Streaming TTS
Total perceived response	~0.5-1.5 s target	Stream partial audio and avoid long answers

The one-second rule in the original document is a useful UX target, not a guaranteed engineering threshold. Real telephone networks and STT/TTS vendors make sub-second end-to-end response difficult. The practical goal is low time-to-first-audio, short responses and streaming. [T1]

10.4 Barge-in

If the farmer begins speaking while TTS is playing, stop the audio output immediately, preserve the conversation state and return to STT. This requires a voice stack with interruption support or explicit audio-stream cancellation.

10.5 Grounding / hallucination control

- The LLM never invents price, queue or payment data.
- All factual answers call an internal tool first.
- If data is missing, the agent says that it cannot verify the value and offers staff assistance.
- Every price response includes the reporting date.
- Every payment response comes from the authoritative status table, not from free-form model memory.

10.6 Current provider pricing

Deepgram's current pricing lists Nova-3 multilingual STT at \$0.0058/minute pay-as-you-go and its voice-agent/streaming products at separate rates; Deepgram's Flux TTS is listed as free through 12 September 2026, after which standard pricing applies. [S15][S16]

Twilio's current India voice page lists India mobile outbound calling at \$0.0496/min and India landline/other outbound at \$0.0699/min. However, Twilio's own India voice guidelines say outbound calls to India can only be made from international, non-Indian numbers. [S17][S18]

That means your earlier assumption of 'Twilio = straightforward production Indian farmer calling' should be treated as a prototype constraint, not a production architecture decision.

11. Telephony and toll-free strategy

11.1 Best SIH approach

For the hackathon, the most reliable path is outbound calling through a telephony configuration that actually supports India calling and the project's compliance constraints, or a simulated call environment for the demo if procurement access is restricted. Do not build the entire project around a phone number you cannot legally or technically provision.

11.2 1800 number

An Indian 1800 toll-free number is feasible in production through Indian cloud-telephony providers, but it is a business/organisation telephony service rather than a free student resource. MyOperator currently advertises 1800 toll-free as an add-on starting at ₹500/month; actual call usage and plan features can add to that. [S19]

A provider guide updated in February/March 2026 states that KYC typically requires business identity/address documents and an authorised signatory; activation can be around 24-72 hours after verification, though exact timelines and documents depend on provider. [S20]

11.3 Toll-free process

1. Choose an Indian cloud-telephony provider that supports 1800 inbound service and API/webhook integration.
2. Complete business KYC / entity verification.
3. Select a standard or vanity 1800 number.
4. Configure call routing to your backend/IVR/AI agent.
5. Configure DLT/SMS compliance separately if SMS is used.
6. Test inbound calls from major Indian operators.
7. Connect call events and recordings/transcripts only with appropriate consent and retention controls.
8. For production government deployment, use the government department/authorised partner as the contracting entity where required.

11.4 Recommended roadmap

Stage	Telephony
SIH prototype	Use a working India-compatible test configuration; outbound demo calls; fallback to controlled simulation if carrier restrictions block the live call.
Pilot	Indian cloud telephony provider + verified organisation KYC + inbound 1800 option if needed.
Production	Official government/partner telephony contract, monitoring, capacity planning, multilingual support, compliance and SLA.

12. AgroChain / Shardeum integration

12.1 Why blockchain belongs here

Blockchain is justified only for events where the team wants a durable, independently verifiable record. It should not replace the operational database.

12.2 Event model

```

PROCUREMENT_CREATED
|
v
QUANTITY_VERIFIED
|
v
PRICE_CONFIRMED
|
v
PROCUREMENT_COMPLETED
|
v
PAYMENT_INITIATED
|
v
PAYMENT_CONFIRMED

```

12.3 What is recorded

- A unique transaction/event identifier.
- Hash of canonicalised transaction fields.
- Event type.
- Timestamp.
- Actor/system identifier (prefer a pseudonymous or system address).
- Previous-event reference where useful.
- Blockchain transaction hash.

The actual personal and financial details remain in Supabase. The blockchain record becomes a proof anchor. This minimises privacy exposure and makes the architecture much easier to defend.

12.4 Major limitation

Blockchain cannot prove that the physical quantity or payment event was honest. It only makes the recorded event harder to alter retrospectively. The system still needs trusted staff processes, authentication, weighing/quality evidence and authorised payment confirmation. This distinction should be stated explicitly to judges.

13. Security, privacy and trust

Risk	Control
Phone number leakage	Role-based access, server-side masking, no public exposure
Aadhaar leakage	Do not store Aadhaar unless absolutely necessary; use authorised identity systems
Bank details leakage	Encrypt/protect, never put on blockchain
AI hallucination	Structured tool calls only
False staff update	Role-based permissions + audit logs
Queue manipulation	Server-side queue rules + immutable audit trail
Blockchain key compromise	Use secure secret storage; separate deployment/admin keys
Replay / duplicate payment event	Idempotency keys
API outage	Cache + stale-data warning
Wrong farmer record	Verified mobile + staff-assisted correction
Call recordings	Explicit retention policy; store only when justified
Prompt injection	Treat external text/data as untrusted; tool allowlist

14. Technical risks and alternative approaches

Problem	Primary approach	Fallback / alternative
Indian voice recognition errors	Deepgram multilingual streaming	BHASHINI/another Indian-language provider benchmark
Voice latency too high	Streaming STT + fast LLM + streaming TTS	Constrain responses to templates for common intents
TTS delay	Flux/Aura streaming	Pre-generate common prompts / use IVR fallback
Government API unavailable	Cache last known daily price	Use official downloadable dataset refresh
Procurement API unavailable	Staff dashboard status entry	CSV/import adapter for pilot
Payment API unavailable	Staff-confirmed status	Reference official system rather than pretending API access
Twilio India outbound limitation	Indian-compliant telephony provider	Controlled demo environment
LLM answer wrong	Tool-first architecture	Template answers for status questions
Blockchain unavailable	Queue events locally	Queue blockchain write asynchronously
Poor network at mandi	Offline-tolerant staff UI	Local cache + later sync
High blockchain cost	Hash-only events	Batch proofs / periodic anchoring
Farmer does not answer	Retry + SMS	Staff follow-up
Farmer distrusts AI	Identify service + source + human fallback	Assisted centre registration

15. Before vs after workflow

15.1 Typical fragmented journey

Farmer hears procurement is open

|

v

Find registration / centre details

|

v

Travel / ask staff

|

v

Wait / discover actual queue

|

v

Sell crop

|

v

Ask when payment will arrive

|

v

Check portal / call / visit again

15.2 KisanCall + AgroChain journey

Register by voice/app

|

v

Receive slot + current govt-reported price

|

v

AI reminder before leaving

|

v

Arrive at centre

|

v

Live queue + ETA

|

v

Procurement recorded

|

v

Proof hash anchored through AgroChain

|

v

Payment status changes

|

v

AI calls farmer

|

v

Farmer can request proof reference

The key benefit is not 'blockchain' by itself. It is that the farmer has a much clearer chain of events and fewer reasons to travel merely to obtain information.

16. Feasibility and cost model

16.1 SIH prototype

Component	Likely SIH cost	Notes
Supabase	\$0	Free tier includes 500 MB DB, 5 GB egress and up to 50k MAU; free projects can pause after inactivity. [S21]
Vercel	\$0	Hobby plan is free for personal/small projects; Pro starts at \$20/month. [S22]
Render	\$0 possible	Free web services available for testing; not recommended for production. [S23]
Deepgram	Potentially low / promotional free credits	Current pricing varies by STT/TTS product; free/promotional terms change. [S15][S16]
LLM	Low / free tier depending on provider/model	Use a fast small model and monitor limits.
Twilio / telephony	Usage based	India mobile outbound price currently listed at \$0.0496/min on Twilio, but India calling restrictions matter. [S17][S18]
Blockchain	Low prototype cost	Use testnet for development and budget only for selected mainnet proof events later.
SMS	Usage based	Use only important alerts to keep cost controlled.
Domain	Optional	Can use platform subdomain during SIH.
Expected SIH operating budget	~₹2,000-₹10,000 sensible reserve	Depends heavily on live call minutes, SMS and provider setup; do not promise an exact ₹500-₹1,500 total.

The original KisanCall document estimated roughly ₹500-₹1,500 for the hackathon. [T1] Current vendor pricing suggests keeping a larger contingency, especially for telephony, and treating free tiers as prototype aids rather than production guarantees.

16.2 Example voice cost

Illustrative example: 100 farmers x 2-minute calls = 200 minutes. At Twilio's current listed India mobile outbound rate of \$0.0496/min, telephony alone would be about \$9.92 before any additional provider/telecom/AI costs. The exact Indian production cost depends on the chosen carrier/provider. [S17]

16.3 Production cost philosophy

At scale, the architecture should be costed as cost-per-completed farmer interaction, not cost-per-feature. The largest variable will be call minutes. Short, tool-driven answers and outbound calls only when valuable can materially control cost.

17. Market progress and impact

17.1 What is happening in India right now

- Government agriculture systems are moving toward interoperable digital public infrastructure and AI-assisted services. [S7][S8]
- The Digital Agriculture Mission has already built large identity and crop-data foundations; a PIB backgrounder in Feb 2026 cited over 7.63 crore Farmer IDs and 23.5 crore crop plots surveyed. [S9]
- Kisan-eMitra demonstrates real farmer usage of multilingual voice AI at government scale. [S6][S9]

- Bharat-VISTAAR signals that voice-first agricultural information access is becoming a national public-infrastructure direction rather than a niche startup idea. [S8]
- Maharashtra's AI-agriculture policy explicitly includes agri-food traceability and farmer-centric AI, which validates the direction toward combined data, AI and traceability systems. [S10]

17.2 Impact model

Metric	How KisanCall can move it	How to measure in pilot
Average waiting time	Pre-schedule arrivals + queue visibility	Compare baseline vs enabled-centre waiting time
Unnecessary trips	Remote status checks	Count trips/calls avoided
Missed slots	Reminder calls/SMS	Missed-slot percentage
Status enquiries to staff	Voice self-service	Calls/questions per 100 farmers
Payment uncertainty	Proactive payment updates	Time-to-notification after status change
Digital inclusion	Voice access	Share of farmers completing core task by voice
Dispute evidence	Transaction proof	Number of events with verifiable record
Farmer satisfaction	Simple post-call survey	CSAT / task completion rate

For SIH, do not claim a percentage reduction without measuring it. Build the dashboard so you can demonstrate those metrics in the pilot.

18. SIH-winning strategy

18.1 The winning positioning

We are not replacing government procurement systems. We are making the existing system accessible, proactive and verifiable for the farmer.

18.2 The one-line USP

KisanCall tells the farmer what is happening; AgroChain proves what happened.

18.3 The three-minute demo

9. Phone rings: KisanCall greets farmer in Hindi.
10. AI states slot and latest government-reported mandi price with date.
11. Farmer asks: 'Mera number kab aayega?'
12. System reads live queue and returns ETA.
13. Staff dashboard advances the queue.
14. Farmer phone receives updated ETA.
15. Staff marks crop verified and procurement completed.
16. AgroChain proof record is created and transaction hash appears.
17. Payment status is changed to PAID in the prototype.
18. AI calls farmer: payment status is explained in natural language.
19. Farmer asks for proof; AI/portal provides transaction reference.

18.4 Questions judges will attack

Likely judge question	Answer you should give
Isn't e-NAM already doing this?	Yes, and we do not claim otherwise. We add a procurement-centre coordination and voice-first farmer experience plus a narrow proof layer.
Why blockchain?	Only key events are anchored for auditability; operational data remains

	off-chain.
Where does payment data come from?	Prototype: staff-authorised status update. Production: authorised government/payment integration.
Is the mandi price truly live?	It is latest government-reported daily price data; we display the date/time and source.
What if the farmer has no smartphone?	Voice-first operation is a core feature.
Why AI instead of IVR?	Natural questions reduce menu complexity; keypad/human fallback remains available.
What if speech recognition fails?	Retry, clarify, keypad fallback and human escalation.
Can this scale?	Stateless backend, shared database, asynchronous proof writes and provider-scale telephony.
Can students deploy an 1800 number?	Not as a casual free hackathon resource. We show the production route and use a working prototype telephony path.
What makes it different?	Integration of procurement coordination, voice accessibility and verifiable transaction events - not any single technology alone.

19. Recommended MVP and four-week execution plan

19.1 Scope control

Do not start with national scale. Choose one crop, one procurement-centre model and one geography. Build a convincing end-to-end flow first.

Week	Build target	Exit criteria
Week 1	Database schema, auth, staff dashboard shell, farmer app shell, price API adapter	Farmer + slot + queue data moves end-to-end
Week 2	Voice pipeline + tool calling + Hindi/English + queue/price/payment intents	Phone demo works for 5-10 scripted conversations
Week 3	Procurement lifecycle + AgroChain proof + SMS + error handling	Full procurement transaction produces proof hash
Week 4	Polish, load tests, failure tests, demo script, metrics	3-minute demo never breaks on rehearsed path

19.2 Minimum database tables

Table	Important fields
farmers	id, name, phone, language, mandi_id, crop
mandis	id, name, district, capacity/day, hours
slots	id, mandi_id, date, start_time, end_time, capacity
bookings	id, farmer_id, slot_id, status, token
queue_events	booking_id, event_type, timestamp, sequence
procurements	booking_id, quantity, price, quality_status, status
payments	procurement_id, status, reference, updated_at
price_cache	mandi, commodity, min, max, modal, date, fetched_at
calls	farmer_id, direction, intent, outcome, duration, timestamp
proof_events	procurement_id, event_type, payload_hash, chain_tx_hash
audit_logs	actor, action, entity, old_value, new_value, timestamp

19.3 Core API endpoints

POST /farmers
 POST /bookings
 GET /farmers/:id/queue
 GET /farmers/:id/status
 GET /mandis/:id/prices
 POST /staff/arrivals
 POST /staff/procurement
 PATCH /payments/:id
 POST /voice/webhook

POST /voice/tool/get-slot
 POST /voice/tool/get-queue
 POST /voice/tool/get-price
 POST /voice/tool/get-payment
 POST /proof-events
 GET /proof/:id

19.4 What to leave for phase two

- Real nationwide government payment API integration.
- All Indian languages.
- Nationwide 1800 service.
- Advanced route optimisation between multiple centres.
- Buyer/trader marketplace expansion from AgroChain.
- Automated dispute resolution.
- Large-scale predictive demand/arrival modelling.

20. Final assessment

The strongest version of this project is not 'AI + blockchain + mandi prices'. It is a carefully bounded digital public-service product:

Government data gives the system facts. KisanCall gives the farmer access. The procurement engine coordinates action. AgroChain anchors proof.

The project is technically feasible for an SIH prototype. The core engineering is ordinary web/mobile/backend work plus a voice pipeline and a limited blockchain integration. The difficult parts are reliability, Indian telephony constraints, language/accent coverage, data provenance, and authorised government integration - all solvable by narrowing the MVP and using fallbacks.

The project is also well timed. Government direction is moving toward voice-first multilingual agriculture services, integrated digital public infrastructure, and AI-enabled farmer services. [S7]-[S10] That makes it strategically smarter to present KisanCall as a specialised procurement interface that complements those systems rather than competing with the entire agricultural digital stack.

Recommended final product name: KisanCall. Position AgroChain as the trust/proof infrastructure, not as a second product the judge has to mentally combine.

Final verdict: 9/10 SIH potential today; 9.5/10 potential with disciplined scope, technically honest integrations, a flawless end-to-end demo, and measurable pilot metrics.

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End of research dossier